EXPERIMENT :- 3

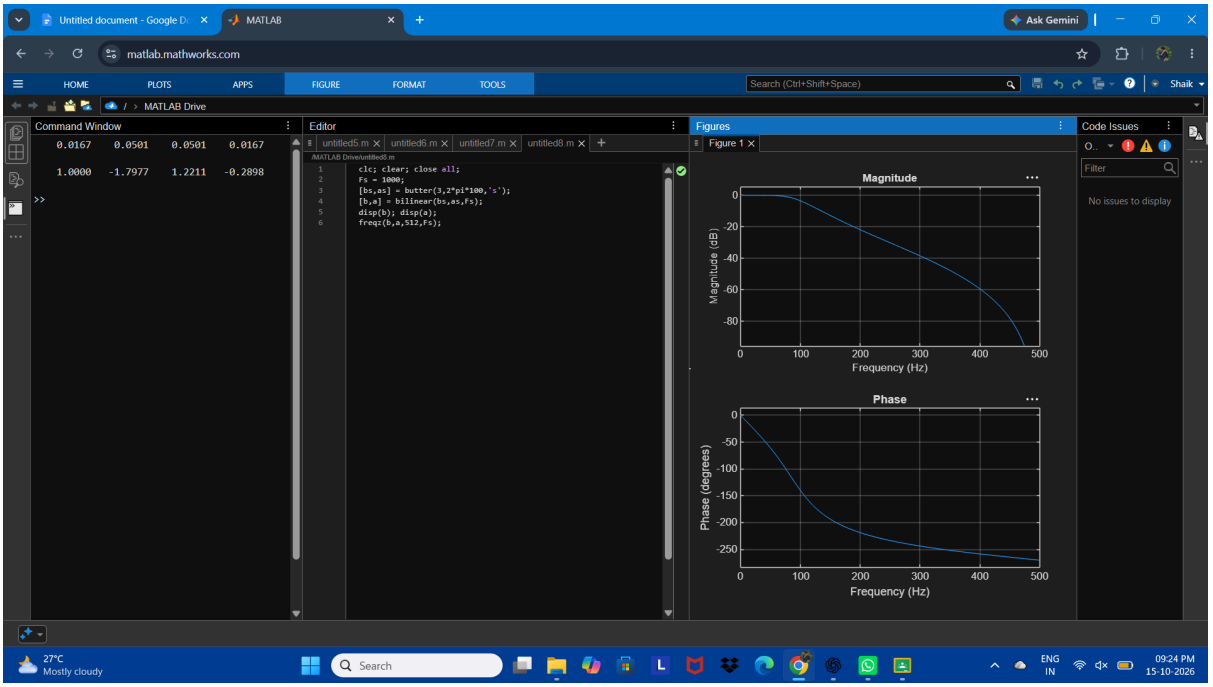
PROGRAM CODES ➖

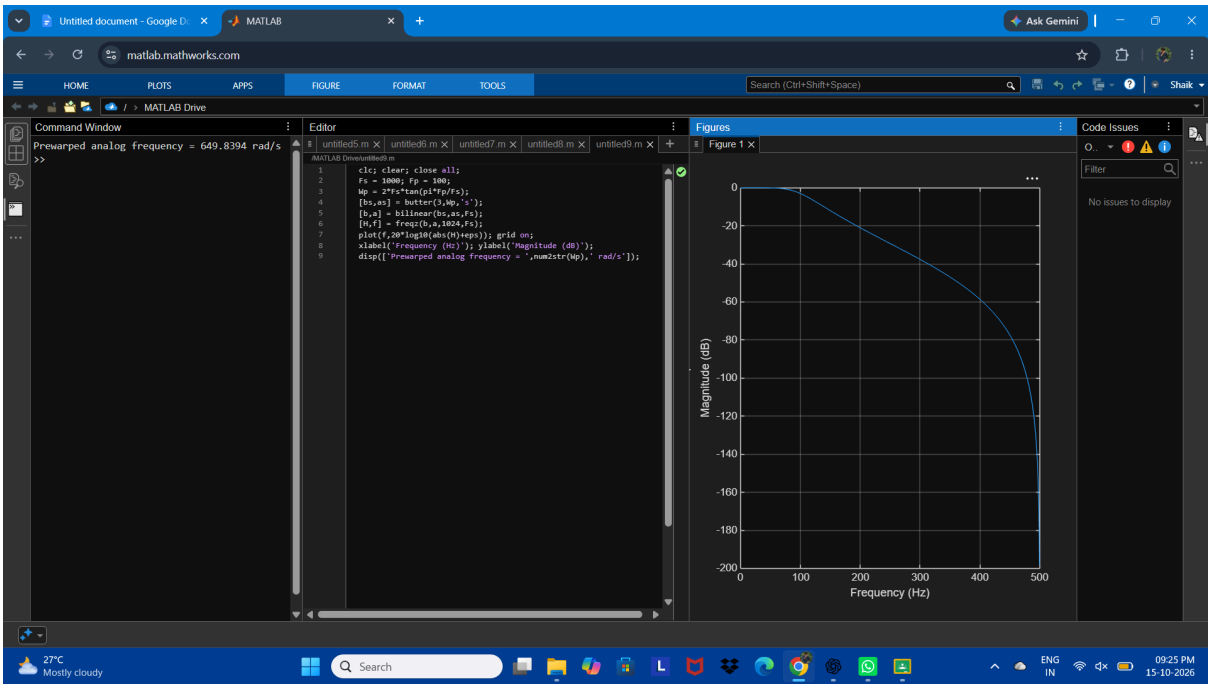
```
clc; clear; close all;
Fs = 1000;
[bs,as] = butter(3,2*pi*100,'s');
[b,a] = impinvar(bs,as,Fs);
disp(b); disp(a);
freqz(b,a,512,Fs);
```

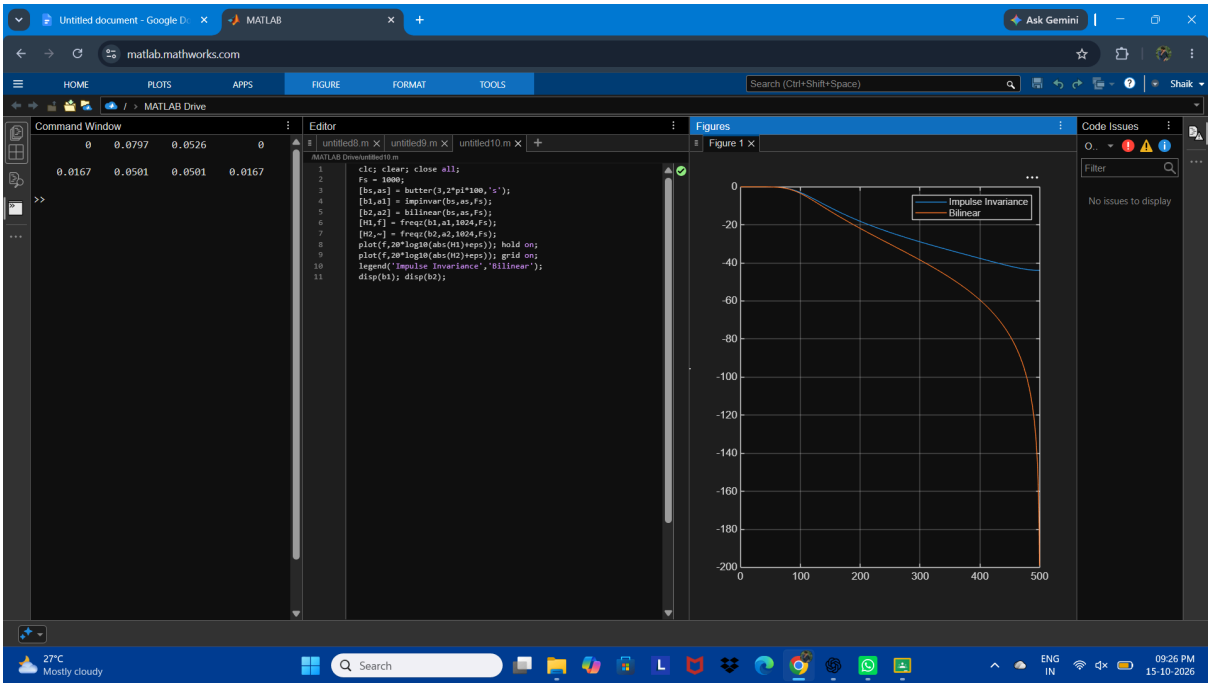
```
clc; clear; close all;
Fs = 1000;
[bs,as] = butter(3,2*pi*100,'s');
[b,a] = bilinear(bs,as,Fs);
disp(b); disp(a);
freqz(b,a,512,Fs);
```

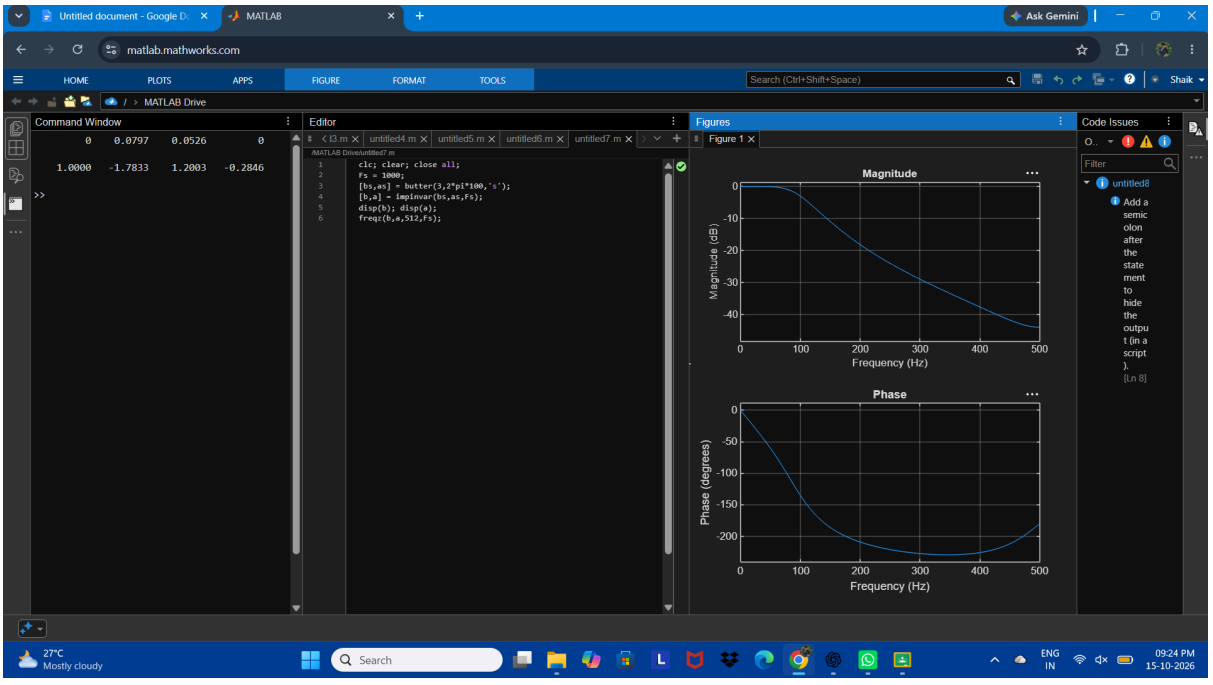
```
clc; clear; close all;
Fs = 1000; Fp = 100;
Wp = 2*Fs*tan(pi*Fp/Fs);
[bs,as] = butter(3,Wp,'s');
[b,a] = bilinear(bs,as,Fs);
[H,f] = freqz(b,a,1024,Fs);
plot(f,20*log10(abs(H)+eps)); grid on;
xlabel('Frequency (Hz)'); ylabel('Magnitude (dB)');
disp(['Prewarped analog frequency = ',num2str(Wp),' rad/s']);
```

```
clc; clear; close all;
Fs = 1000;
[bs,as] = butter(3,2*pi*100,'s');
[b1,a1] = impinvar(bs,as,Fs);
[b2,a2] = bilinear(bs,as,Fs);
[H1,f] = freqz(b1,a1,1024,Fs);
[H2,~] = freqz(b2,a2,1024,Fs);
plot(f,20*log10(abs(H1)+eps)); hold on;
plot(f,20*log10(abs(H2)+eps)); grid on;
legend('Impulse Invariance','Bilinear');
disp(b1); disp(b2);
```


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clc; clear; close all;
Fs = 1000;
[bs,as] = butter(3,2*pi*100,'s');
[b,a] = bilinear(bs,as,Fs);
disp(b); disp(a);
freqz(b,a,512,Fs);
Figures
Figure 1
Magnitude
Magnitude (dB)
Frequency (Hz)
Phase
Phase (degrees)
Frequency (Hz)
Code Issues
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clc; clear; close all;
Fs = 1000; fp = 100;
Wp = 2*Fs*tan(pi*fp/Fs);
[bs,as] = butter(3,Wp,'s');
[b,a] = bilinear(bs,as,Fs);
[H,f] = freqz(b,a,1024,Fs);
plot(f,20*log10(abs(H)+eps)); grid on;
xlabel('Frequency (Hz)'); ylabel('Magnitude (dB)');
disp(['Prewarped analog frequency = ',num2str(Wp),' rad/s']);
Figures
Figure 1
Magnitude (dB)
Frequency (Hz)
Code Issues
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clc; clear; close all;
Fs = 1000;
[bs,as] = butter(3,2*pi*100,'s');
[b1,a1] = impinvar(bs,as,Fs);
[b2,a2] = bilinear(bs,as,Fs);
[H1,f] = freqz(b1,a1,1024,Fs);
[H2,~] = freqz(b2,a2,1024,Fs);
plot(f,20*log10(abs(H1)+eps)); hold on;
plot(f,20*log10(abs(H2)+eps)); grid on;
legend('Impulse Invariance','Bilinear');
disp(b1); disp(b2);
Figures
Figure 1
Impulse Invariance
Bilinear
Code Issues
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clc; clear; close all;
Fs = 1000;
[bs,as] = butter(3,2*pi*100,'s');
[b,a] = impinvar(bs,as,Fs);
disp(b); disp(a);
freqz(b,a,512,Fs);
Figures
Figure 1
Magnitude
Magnitude (dB)
Frequency (Hz)
Phase
Phase (degrees)
Frequency (Hz)
Code Issues
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Add a semicolon after the statement to hide the output (in a script).
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